

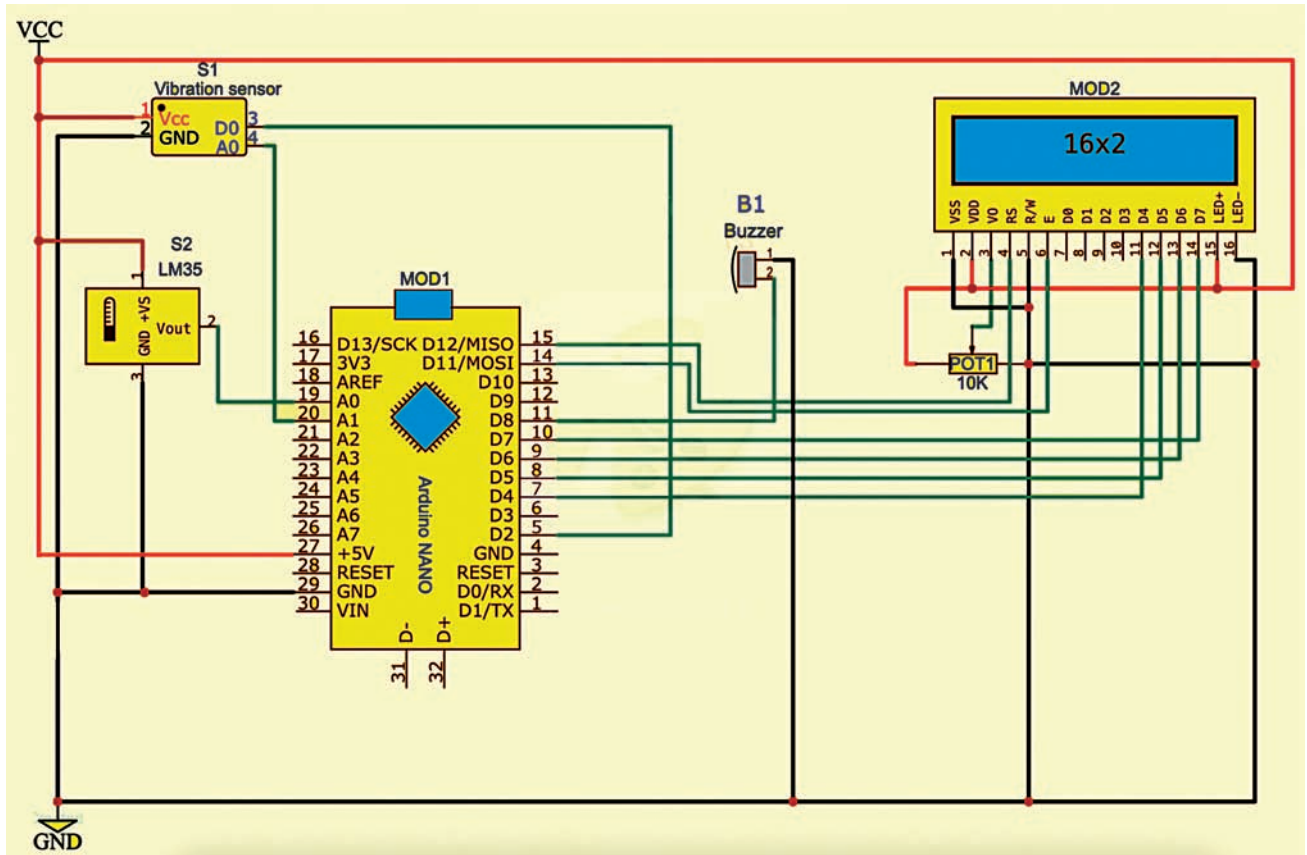


Fig. 2: Circuit diagram

a 2-Wire I2C LCD display can also be used.

Software

The software is based on the Arduino IDE platform and consists of three separate programs:

The vibration monitoring program measures and plots vibration amplitude using the serial monitor and plotter, helping lathe machine operators identify excessive vibrations caused by misalignment or mechanical issues.

The temperature monitoring program monitors lathe temperature and triggers an alert if it rises beyond the set threshold. Alternatively, a relay with a driver transistor can be added after modification to shut down the lathe at extreme temperatures or activate a cooling system through a third program.

```
#define sensorPin A0 // Analog pin connected to the sensor (for vibration amplitude)
#define digitalPin 2 // Digital pin connected to the sensor (for vibration detection)

void setup() {
  Serial.begin(9600); // Start serial communication at 9600 baud rate
  pinMode(sensorPin, INPUT); // Set the sensor pin as input
  pinMode(digitalPin, INPUT); // Set the digital pin as input
}

void loop() {
  int digitalValue = digitalRead(digitalPin);
  if (digitalValue == HIGH) { // Vibration detected
    int analogValue = analogRead(sensorPin);
    float voltage = analogValue * (5.0 / 1023.0);
    float vibrationSpeed = voltage * 20.0;
    Serial.print("Vibration Speed: ");
    Serial.print(vibrationSpeed);
    Serial.println(" units");
  } else {
    Serial.println("No Vibration Detected");
  }
  delay(500);
}
```

Fig. 3: Snippet source code for the vibration monitoring program

These programs can be used separately or combined into a single functional system.

To use the vibration monitoring program, upload the vibration.ino source code to the Arduino Nano. The

serial monitor and plotter will then display vibration data in real time.

Construction and testing

Upload the source code to the Arduino Nano using the appropriate